AI Film Production Acceleration Plan

A concise, end-to-end blueprint to accelerate AI film production from brief to deliverables.

1. Capture and organize project briefs efficiently
- **Design structured brief intake**
  - Build a form (web or CLI) capturing logline, genre, tone, target audience, runtime, constraints (voice type, stock limits), and delivery specs.
  - Store briefs as versioned JSON/YAML documents with unique project IDs.
  - Add validation (required fields, length limits) and templates for common genres (ad, explainer, short film).

2. Convert brief into high-level concept and beat outline
- **Automate concept and beat-sheet generation**
  - Prompt an LLM with the brief JSON plus few-shot examples to produce premise, theme, visual motif, and 8â 12 beats.
  - Enforce schema via JSON output (beats with description, visual idea, duration, emotional tone).
  - Add a â regenerate by sectionâ control so users can reroll specific beats without losing the rest.

3. Produce full script with dialog and stage directions
- **Script generation workflow**
  - Feed beats into an LLM prompt to expand into a script (sluglines, action, dialog) using a screenplay format (e.g., Fountain/Final Draftâ like plain text).
  - Include controls for pacing (target runtime per scene), reading level, and brand/legal constraints (no trademarks, safe content).
  - Store script revisions with diff/compare view and per-scene regenerate.

4. Turn script into storyboard frames
- **Storyboard frame extraction**
  - Parse script into scenes and shots; extract characters, location, time of day, and camera notes (heuristics or LLM extraction).
  - Generate shot cards (JSON) containing shot type, composition, lens, movement, lighting, mood, and key props.
  - Render to a visual storyboard (static HTML/PDF) and allow manual edits per card.

5. Generate image prompts for each shot
- **Image prompt builder**
  - Map shot card fields to prompt templates (e.g., ‘${style} ${lens} ${lighting} ${composition} of ${subject} at ${location}‘).
  - Support multiple target models (SDXL, Midjourney, DALL´•E) with style presets and negative prompts.
  - Add safety filters (ban lists) and aspect-ratio selection tied to delivery format (9:16, 16:9, 1:1).

6. Create voiceover and timing guides
- **Voiceover + timing pass**
  - Use TTS to generate scratch VO from script and capture timestamps.
  - Derive timing marks per line/beat; export as SRT/VTT for alignment with visuals.
  - Allow alt takes (speed/pitch) and a timing adjustment slider to nudge cues earlier or later.

7. Auto-generate video assembly prompts and edits
- **Rough-cut assembly**
  - Combine storyboard images (or generated stills), VO, and music bed into a rough cut via an ffmpeg pipeline; auto-fit durations to VO timing.
  - Generate LLM-assisted edit lists (EDL) describing transitions, overlays, and motion (pan/zoom) and render a draft MP4.
  - Export a structured edit timeline (e.g., OpenTimelineIO/EDL/JSON) for NLE import (Premiere/Resolve).

8. Graphics and on-screen text generation
- **Graphics/text automation**
  - Extract lower-third, supers, captions, and CTA requirements from script/brief.
  - Generate styled SVG/PNG assets using brand palettes and typography presets.
  - Place overlays onto the timeline JSON with in/out points and safe-title margins.

9. Feedback loop and versioning
- **Review and iteration tools**
  - Provide per-stage checkpoints (brief â beats â script â storyboard â prompts â rough cut) with approvals.
  - Track versions with tags and change logs; allow â forkâ to explore alternates.
  - Add inline commenting on frames/shots and regenerate buttons scoped to the commented item.

10. Delivery and export
- **Packaging outputs**
  - Export bundles: script (PDF/Markdown), storyboard (PDF/HTML), prompt pack (CSV/JSON), VO (WAV), EDL/OTIO, and draft video (MP4).
  - Include a manifest JSON summarizing assets, durations, and model settings used.
  - Add a publish step to upload to storage (S3/GCS) with shareable links.

11. Tech stack and infrastructure
- **Stack selection and scaffolding**
  - Choose a web stack (e.g., Next.js/TypeScript) or CLI (Python) plus a backend service (FastAPI/Node) for orchestration.
  - Add worker queue for long-running tasks (Celery/RQ/BullMQ) and object storage for assets.
  - Wrap LLM/image/voice calls with provider abstractions to swap models easily; log prompts/responses for traceability.

12. Safety, compliance, and guardrails
- **Guardrails and safety checks**
  - Pre-filter briefs and prompts for disallowed content; enforce negative prompts and trademark filtering.
  - Add per-output checks: hallucination guard on facts, NSFW detectors on images, and profanity checks on VO/scripts.
  - Keep an audit log of model calls and user overrides.

13. UX considerations
- **User flow design**
  - Create a wizard-like flow with save/resume and progress indicators for each stage.
  - Provide inline preview of generated frames, VO playback, and edit timeline scrubber.
  - Offer smart defaults and a â one-click draftâ that runs the entire chain, then surfaces sections to refine.

14. Evaluation and quality
- **Quality scoring and A/B tests**
  - Add rubric scoring (story clarity, pacing, visual coherence) via LLM and simple heuristics.
  - Allow A/B generations (two styles/cuts) and collect user preference feedback.
  - Track metrics such as turnaround time, regenerate rate per step, and export success.